Ng Yu Hang (Mervin)

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EDUCATION

Singapore Polytechnic Singapore

Diploma in Applied AI & Analytics; CGPA 3.83 / 4.00 *Apr 2024 - Apr 2027 (expected)*

- Director's Honour Roll AY25/26; top 5% in Deep Learning, Practical AI, DevOps for AI

EXPERIENCE

Founding Product & AI Engineer (Part-time) Sep 2026 - Present

Influencees *Singapore (Hybrid)*

- Built Ai-kyo, an agentic AI assistant for creators, by orchestrating an intent-routed LLM agent with tool integrations
- Unified 6 messaging channels into one inbox (WhatsApp, Instagram, LinkedIn, TikTok, Gmail and more)
- Cut inbox sign-in lag from 1.1 s to 12 ms by profiling the streaming path; shipped 10 PRs to production

Full-Stack AI Engineer (Intern) Mar 2026 - Present

SP Data Science & Analytics Centre *Singapore*

- Cut retrieval time ~70-85% and manual search ~60% for EM Services with an Azure AI Search RAG chatbot
- Cut Power BI grading time ~50-70% with an LLM grader that scores .pbix dashboards against rubrics

AI Researcher / Developer (Part-time) May 2026 - Aug 2026

PetaniAI *Singapore (Hybrid)*

- Informed model choices by benchmarking LLMs on the company's own requirements

Student Developer Sep 2025 - Feb 2026

Singapore Polytechnic, School of Computing *Singapore*

- Delivered a user-tested GenAI chatbot for a paid client; presented to SP's Deputy Principal and PCEO

PROJECTS

AgentLabs | *Python, FastAPI, React, PyTorch (LoRA), Docker* 2026

- Hardened AI agents against 200 adversarial attacks via an attack, verify, auto-patch and retest loop (live demo)

ProofBench | *Python, FastAPI, PostgreSQL, Docker, GitHub Actions* 2026

- Evaluated developer tools by generating adapters, running them in sandboxes and scoring output deterministically

Lactic | *Python, Flutter, Supabase, embeddings* 2026

- Built an A-Level AI tutor with hybrid vector and graph retrieval over 31k+ chunks

Edge Human Detection | *Python, PyTorch, YOLO, OpenCV, Docker* 2026

- Fine-tuned YOLO11 on thermal imagery for night-vision person detection with multi-frame checks

ACHIEVEMENTS

Projet: Hack & Hire: Winner

'SUP Build2026: Top 3 + Best Use of Zo

Daytona HackSprint: 2nd runner-up

NAISC (Huawei Track): Top 10

National Olympiad in AI 2026: Bronze (top 60)

OpenClaw Hackathon: Top 3

Agent Forge Hackathon: 3rd runner-up

PolyFinTech 2026 (ESG): Finalist

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Dart, Go
Generative AI	RAG, agents, tool use, LLM evaluation, LoRA fine-tuning, prompt engineering, VLMs
ML	PyTorch, TensorFlow, Hugging Face, scikit-learn, LangChain, YOLO, OpenCV
Engineering	FastAPI, Node.js, React, Next.js, Docker, Azure, AWS, PostgreSQL, Supabase, GitHub Actions
Certifications	AWS Certified AI Practitioner, Google AI Professional, AI Singapore Certified Trainer